Expertise

Introductory article

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Expertise refers to the cognitive, perceptual-motor, and physiological mechanisms that allow experts to attain consistently superior levels of performance on representative activities in their domains.

EXPERTISE AND EXPERT PERFORMANCE

In everyday terminology, experts are individuals, such as medical doctors, accountants, teachers and scientists, who have first been certified as professionals after extended training and then have gained substantial experience in their specialty. More recently, the term has been expanded to describe any highly skilled performer who consistently exhibits superior achievement after instruction and extended experience in a domain such as one of the arts (e.g. music, painting, writing), a sport (e.g. swimming, soccer, golf) or a game (e.g. chess, Othello, bridge). The general concept of expertise is based on the hypothesis that experts in the different domains will acquire their superior performance according to similar learning principles and that the general structure of the acquired mechanisms will be similar in domains with similar demands on performance.

The main task for researchers of expertise is to explain how some individuals attain the highest levels of achievement in a domain, and why so few reach such levels. Expert performance in many domains looks effortless, and one is thus led to believe that experts excel in general basic characteristics, such as intelligence, memory, speed, and flexibility. It has been assumed traditionally that such characteristics are impossible to train and thus are determined to a large degree by genetic factors (nature). Everyone agrees, however, that experts must acquire at least some necessary domain-specific knowledge and skill (nurture). Without denying the possibility of effects of genetic factors, research in expertise tends to emphasize the importance of knowledge and acquired skills.

Sir Francis Galton, a pioneer in the study of excellence in nineteenth-century England, claimed that instruction and training were beneficial, even necessary, and associated with large initial improvements in performance. However, he also maintained that the upper bound of individuals' performance was fixed by hereditary factors. This view is still common among many researchers and practitioners in some domains. Thus, proponents of this view are still searching for innate talent and measuring basic capacities of memory, perception, and thinking, that would allow them to successfully predict future outstanding performance in children and untrained adults. Today, over a century later, we can safely say that efforts to measure individual differences in basic capacities have not succeeded in predicting which individuals would eventually attain expert performance. For example, when athletes or other experts are tested in the laboratory on how fast they can respond to the onset of a light (simple reaction time), they are not systematically faster than other people. The amazing speed of tennis players' actions is thus not due to superior speed of neural impulses, and we have to assume that returning a fast tennis serve reflects an acquired ability to respond rapidly in specific types of situations.

Domain-specific superiority of experts is very noticeable for cognitive abilities. Chess masters can recall nearly all the 24 chess pieces in a typical chess arrangement after a quick glance, whereas beginners in chess can recall only around four pieces. The experts' ability to recall more chess pieces is, however, restricted to representative (or typical) chess positions in which the expert can meaningfully encode the relations between the chess pieces. When random arrangements of chess pieces are presented, the large recall advantage of the experts over the novices is virtually eliminated.

The major differences between experts and less proficient individuals nearly always reflect specific adaptations acquired by the experts during their lengthy training. This holds true for many anatomical and physiological characteristics of athletes, such as the size of their muscles and bones and the flexibility of joints, and for the increased range of mobility of the limbs in ballet dancers and musicians. Some of these attributes, such as structural changes in the brains of musicians, are even correlated with the early onset of training. Other adaptations – for example, the growth of muscles and thickening of the bones of tennis players (restricted to the arm holding the racket), or the optimization of metabolism of runners (for particular running speeds and different lengths of races) – are so specific that (self)-selection of individuals in those domains appears unlikely. Finally, many physiological adaptations, such as the heart size of endurance runners, have been shown to revert to the normal values once the athletes stop training, which would be expected only for acquired and not for innate characteristics. However, there is at least one exceptional attribute – height – where genetic factors are known to control the development of body size. Above-average height constitutes an advantage in attaining the highest levels in some sports such as basketball, but is a disadvantage in other sports such as gymnastics.

The occurrence of expert achievement in similar domains across several generations in some famous families is frequently cited as proof for the genetic transmission (high heritability) of special talents. However, this can be questioned since the early instruction of children by parents and access to networks and specialized training seem to offer plausible alternative accounts. So far, studies on experts have been unable to document any convincing evidence for significant heritability of expert levels of achievement.

EXPERTISE AS ACQUIRED KNOWLEDGE AND SKILL

Organized Knowledge as a Base for Expertise

Our understanding of expertise was advanced primarily by comparing the thought processes of experts while exhibiting their superior performance with the thoughts of less accomplished individuals (or even novices) performing the same tasks. In his 1940s pioneering study, researcher Adrian de Groot wanted to explain why world-class players generated consistently better chess moves than less skilled players after only a brief exposure to the chess position. He instructed chess players in each group to think aloud while they selected their next moves for a given chess position. The transcriptions of these reports (think-aloud protocols) revealed that in an initial phase all chess players were quickly generating promising moves while examining the organization and structure of the presented chess position. Next, the chess players spent minutes evaluating these potential moves by searching and planning and discovered often even better moves. Neither de Groot nor other researchers found that world-class players differed from less skilled chess players with regard to basic intelligence or to speed of their planning. The rapid speed of move generation indicated that the superior moves were directly retrieved from knowledge of similar chess positions in memory.

According to Simon and Chase's theory of expertise, the essential factor is the accumulation of increasingly complex patterns that allow experts to maintain access to their vast body of knowledge and experience. The same mechanisms were proposed to explain the efficient problem-solving and decision-making of experts in physics, medicine, and business. Hence, the superior ability of experts to reason was found to be specific to domain-related material and thus depended directly on their large body of knowledge and experience. Viewing expertise in terms of superior knowledge led investigators in artificial intelligence to interview experts in order to extract the necessary relevant knowledge and decision rules for building computer programs (expert systems) that were designed to reproduce the experts' behavior.

Challenges to the Knowledge-based View of Expertise

Paradoxically, individuals recognized as domain experts because of their lengthy training and experience do not always exhibit consistently superior performance compared with less experienced individuals and sometimes even complete amateurs. For example, professional stockbrokers do not consistently select better investments than statistical models and amateurs, or even random picks. Impressive dissociations between the level of expertise (indicated by the amount of schooling and experience) and performance have been demonstrated in medical diagnosis of common diseases, treatment with psychotherapy, auditing, and many types of expert decision-making and forecasting.

Why does extended experience not always lead to improvements in performance? It is well known

that when persons are introduced to an activity, their gains in performance are initially large, but once an acceptable level of performance is reached, the behavior of individuals tends to become stable and automated with no further improvements of performance. For example, the vast majority of amateur musicians and recreational athletes in golf and tennis remain at a stable level of performance for years and decades despite their regular involvement in relevant activities. Therefore, it is necessary to distinguish individuals who are customarily labeled 'experts' because of their extended experience, training and reputation, from true 'expert performers' who exhibit consistently superior performance.

Superior Achievement as a Prerequisite for Expert Performance

If we restrict expertise to reproducible expert performance, then the first step is to design tasks that can reliably measure the associated superior performance under standardized conditions. In sports, athletes compete under fair conditions, when they run the 100 m dash or swim 200 m freestyle. Competitions in music and ice skating rely on panels of judges to rate performance. In chess and tennis, the best performers are identified by tournaments where winners advance. In most domains procedures have evolved to evaluate and measure performance in order to identify individuals with superior performance.

The Necessity for Experience, Instruction, and Practice

The development of expert performance requires extensive experience, and it progresses gradually without sudden increases in performance level over extended periods. This is true for the development of adult experts and even for child prodigies. Most expert performers continue to improve their performance long into adulthood, and attain their peak performances in vigorous sports typically at age 25–35 years, and much later for achievements in the arts and sciences. Simon and Chase showed that it takes at least 10 years of training and experience in chess even for the most talented individuals to reach an international level after the start of their engagement in the domain. This 10-year rule has since been shown to apply to most other domains, even in swimming and music where children start practice as early as age 4 or 5 years. Extended experience is thus necessary but not sufficient to attain high levels of performance. If mere experience does not automatically change individuals' behavior, what does?

When our actions and activities run smoothly, a change in the structure of performance is generally not necessary and hence would not be expected. Individuals tend to think that their performance in a given area is 'good enough', and mistakes occur only for very rare or difficult behavior or decisions. For example, a tennis player might have problems with a particular backhand volley, but the need for this stroke during recreational play might occur rarely and unpredictably. In these typical circumstances more experience would not necessarily be associated with improvement of the stroke. However, the problem could be remedied with some precise suggestions by a coach and subsequent practice. The coach would perhaps start with preparatory exercises and work up to difficult and unpredictable exchanges, provide enough challenge and repetitions, and eventually incorporate the strokes in regular training matches.

This optimal type of training activity has been called 'deliberate practice', and its sole purpose is to improve performance effectively. Note that the particular activities that constitute deliberate practice will vary from domain to domain and differ with the level of attained skill. Recreational athletes and amateurs rarely engage in deliberate practice, although they realize that deliberate practice would improve their performance. Since engaging in a domain activity is usually motivated by inherent enjoyment (play) or external rewards (work), most individuals shy away from activities that are more effortful or less enjoyable than the regular activity. Consequently, regular recreational activities and work lack the prerequisites of deliberate practice that are essential for efficient improvement, namely specific goals for training, feedback, and opportunities for gradual improvement through repetition. Engagement in deliberate practice appears to be imperative for individuals to develop the mechanisms for reaching the highest levels of performance.

Over time, experts in domains of expertise have extracted and accumulated a body of organized experience in the form of knowledge and skills that can be effectively transmitted by teachers to students with developed practice techniques. For example, in the Middle Ages it was believed that 50 years were required to master the known forms of mathematics, whereas today students acquire comparable knowledge in highly organized form during their high-school years. Similarly, improved training allows serious amateurs in sports events such as running or swimming to reach levels of performance attained only by elite performers

earlier in the twentieth century. Instead of having to rediscover items of knowledge and ways of practicing or learning, individuals can be given instruction to master and even surpass the established levels of attainment.

The need for specialized instruction and deliberate practice has been recognized. In many domains master teachers and coaches instruct and supervise promising individuals and design their practice from a very young age. The core assumption of deliberate practice is that expert performance is acquired gradually and that effective improvement of the student's performance depends on the teacher's ability to isolate a sequence of simple training tasks that the student can then successively master by repetition with feedback and instruction. The individual training tasks have to be difficult enough to lie slightly outside the student's current range of skills so that the student concentrates on critical aspects and gradually refines performance through repetition in response to feedback. This requirement of focused attention on individual task components differentiates deliberate practice from both mindless drill and playful engagement.

Successful expert performers in different domains share some biographical characteristics, and Bloom and his colleagues have developed a schematic model of the phases of skill acquisition (Figure 1). Often, future experts start training with teachers at young ages after a brief period of playful interaction with the domain (phase I). Their initial formal training (phase II) consists of short training units of about 15–20 min per day, and often a parent supervises practice and helps the child to concentrate. With increasing age, domain-related activities, especially deliberate practice, occupy more and more room in the daily lives of future expert performers, until at the end of adolescence their commitment to the domain is essentially full-time (phase III). During this phase, increases in performance go hand in hand with the acquisition of the cognitive mechanisms that mediate the superior performance (see below). Bloom found that to reach an international level of achievement, the performers – almost without exception – had been trained by excellent teachers who either were international level experts themselves or had successfully trained students to reach international levels.

At some point the experts start their professional career (phase IV). During this last phase the elite performers strive to make eminent contributions to the domain, which could constitute the setting of new world records in sports, publishing scientific discoveries, or making major creative contributions to the arts.

By using diaries and other methods to study how expert musicians spent their daily lives throughout the different phases, Ericsson and his colleagues demonstrated the importance of deliberate practice for attaining expert performance. They investigated three groups of experts differing in their level of attained musical performance. Based on retrospective estimates of past practice times, the researchers calculated the number of hours of deliberate practice accumulated by the different groups of musicians (Figure 2), and found that the better musicians had spent more time in deliberate practice during their development. By the age of 20 years, the best piano players had spent over 10 000 h in practice compared with only around 2000 h for typical amateur pianists of the same age. A number of subsequent studies in chess, sports, and music have confirmed the relationship between the level of attained performance and amount or quality of deliberate practice.

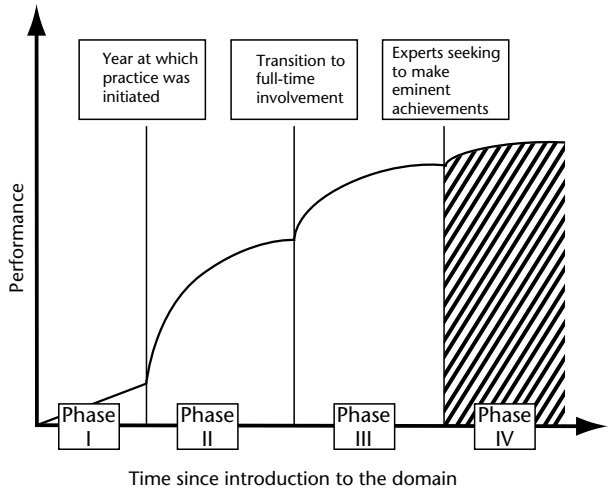


Figure 1. Bloom's three phases of acquisition of expert performance, followed by a qualitatively different fourth phase during which experts attempt to go beyond the available knowledge in the domain. From Ericsson KA, Krampe RT and Heizmann S (1993) *The Origins and Development of High Ability*, pp. 222–249. Chichester, UK: John Wiley. Copyright 1993 by CIBA Foundation. Adapted with permission.

EXPERT PERFORMANCE

The extended period of deliberate practice enables expert performers to acquire complex mental mechanisms that mediate superior performance in their domains. These same mental mechanisms allow them to become their own teachers, which in turn permits them to continue learning and

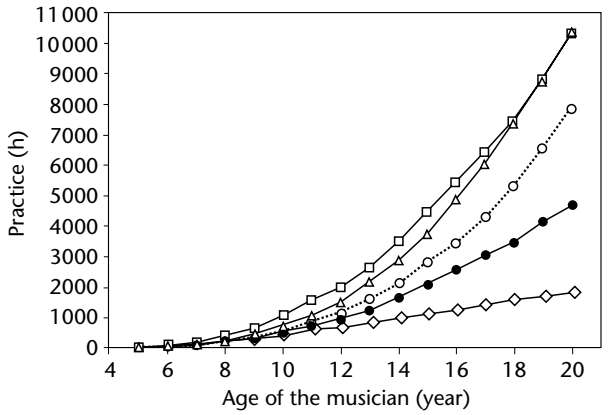


Figure 2. Amount of time spent in solitary practice as a function of age for the middle-aged professional violinists (triangles), the best expert violinists (squares), good expert violinists (open circles), the least accomplished expert violinists (solid circles) and amateur pianists (diamonds). From Ericsson KA, Krampe RT and Tesch-Römer C (1993) The role of deliberate practice in the acquisition of expert performance. *Psychological Review* **100**(3): 379, 384. Copyright 1993 by American Psychological Association. Adapted with permission.

improving their performance for their entire professional career.

Cognitive Mechanisms Underlying Captured Expert Performance

In most domains it is possible to identify the representative tasks and cognitive mechanisms that capture the essence of that expertise. For example, the essence of chess expertise is the consistent ability to select the best move on the chessboard. Hence, following de Groot's paradigm, we can present different chess players with the same series of unfamiliar chess positions and ask them to think aloud as they select the best move (Figure 3(a)). By analyzing the think-aloud protocols, we can scientifically investigate the cognitive processes that differentiate the experts' performances from those of less skilled players.

In chess, analyses of such think-aloud protocols reveal that chess experts rapidly encode the chess position and access adequate moves from memory. However, chess experts will then spend several minutes evaluating these potential moves by planning out the consequences for each of them. The ability to predict the consequences a particular chess move would have several moves ahead requires extensive working memory. Chess players acquire memory skills that allow them to store information about planned chess positions in long-term working memory (LTWM). The proficiency of LTWM increases slowly as a function of chess skill and can account for increases in the ability to plan, as well as the previously discussed increased memory for briefly presented chess positions. Although chess players can often select appropriate moves very early in the decision process, with additional time the quality of the selected moves will increase for all levels of chess players. Thus, at higher levels of performance, experts have acquired the ability to mentally represent relevant information. Chess masters, for instance, have perfected their mental representations of the chess positions to the point where they can play chess blindfolded.

Mental representations have been shown to play a central role in all domains where it has been possible to capture expert performance with laboratory tasks. For example, medical expertise can be captured by presenting doctors with information about challenging medical cases and asking them to reach a diagnosis (Figure 3(b)). From analyses of protocols of medical experts and students thinking aloud while diagnosing patients, it is known that the experts have acquired a superior LTWM with a better representation of the relevant clinical information. As a result, to arrive at the correct diagnosis medical experts are able to plan and reason about the many relevant symptoms.

In domains of perceptual-motor activity, acquired mental representations allow experts to anticipate upcoming events. For example, the best predictor of individuals' keyboard typing speed is not the basic speed of their finger movements but how far they look ahead. By planning ahead, expert typists can anticipate future keystrokes and thus move fingers to their positions ahead of time rather than rush their fingers. Similarly, the rapid reactions of athletes, such as hockey goalkeepers, baseball hitters, and tennis and football players, have also been found to reflect the ability to anticipate events. The resulting shorter reaction times of experts (compared with less accomplished individuals or novices) for representative tasks are thus not due to an innate speed advantage but to superior anticipation, preparation, and improved perceptual skills.

In many domains, such as music, dance and some sports, the key to expert performance is the acquisition of representations that enable the performer to produce the same movement repeatedly. For example, superior golf players can execute the same putt several times with fewer variations than less skilled golfers; similarly, expert pianists are better able to play the same musical piece several

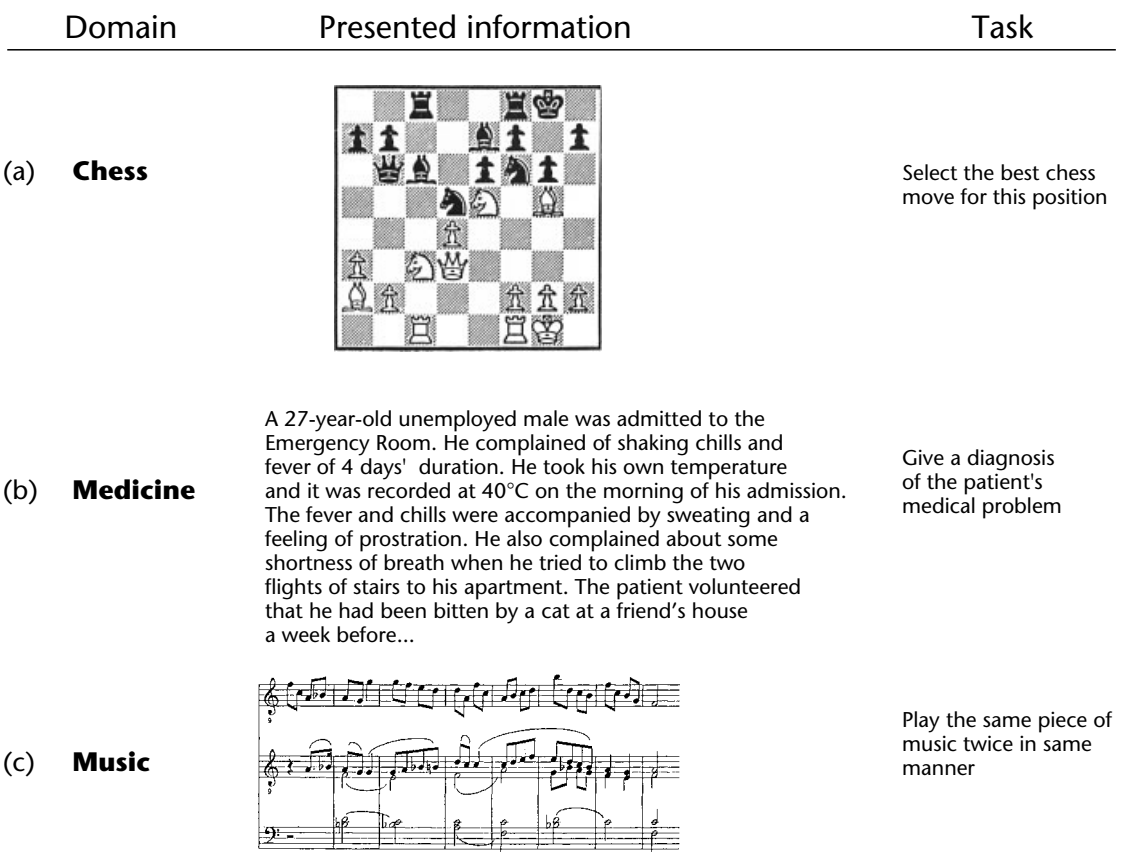


Figure 3. Three examples of laboratory tasks that capture the consistently superior performance of domain experts in chess (a), medicine (b), and music (c). The description of the medical case was taken from the first part of a case report of acute bacterial endocarditis, from 'Inferences in clinical case comprehension' by LD Coughlin and VL Patel, in *Technical Report in Cognitive Research Series*, Center of Medical Education, McGill University, Montreal, Canada, p. 21, 1987. Adapted from Ericsson KA (1998) The scientific study of expert levels of performance: general implications for optimal learning and creativity. *High Ability Studies* **9**: 92. Copyright 1998 by European Council for High Ability.

times in a consistent manner than less skilled pianists (Figure 3(c)).

In sum, the experts' performance is not characterized by reduced cognitive processing and automaticity; rather, experts acquire and refine their mental representations to increase control over their performance.

Importance of Mental Representation for Continued Learning

The experts' ability to mentally represent and manipulate relevant information endows them with the level of control that they need to respond flexibly to changing conditions in real life. Imagine a pianist who has to play a well-entrenched piece under changing acoustical conditions and on different instruments every night, or a tennis player who needs to adapt to different opponents' strengths and weaknesses. In order to maintain a high level of performance despite changes in the environment, the expert needs a flexible and generalizable skill.

Reaching expert performance in a domain does not simply consist of gradually increasing the level of performance, but rather requires the acquisition and the refinement of underlying mental representations, such as being able to image and plan a desired performance, and monitor and evaluate one's own ongoing performance. These are the ultimate goals of instruction, because they make the learner independent of teachers and coaches. In fact, the learner becomes his or her own teacher or coach. This ability to become one's own teacher is important for the development of domain-specific skills at a societal level. As mentioned earlier, the primary objective of experts in later stages of their lives is to make a personal creative contribution to the domain. Before making an innovative contribution to a respective domain, the

performer has to master all the available knowledge and skill; otherwise, he or she would not know when and how to go beyond the current knowledge of performance.

How do experts use their highly developed representations to organize their own training and further improvement in performance? A common method involves studying and analyzing performances and achievements of masters in their respective domains. For example, expert chess players collect books and magazines with published games of chess masters and spend several hours every day playing through those games move by move, trying to predict the next best move the master could have chosen. Any inconsistency between their own prediction and the chess master's actual move is carefully analyzed for oversights in planning and evaluation. In general, this form of study is theoretically interesting because attempting to copy the model behavior of established masters allows performers to gradually refine their own independent representations and expand their body of knowledge. Other domains are likely to require other types of activities that promote further improvements and innovations.

CONCLUSION

The public performance of expert musicians and dancers often appears to be effortless and natural. However, research on expert performance shows that even the most 'talented' individuals need around 10 years and thousands of hours of deliberate practice to attain an international level of performance in their respective domains. Research that has successfully captured expert performance with representative tasks in the laboratory has revealed the complexity and domain specificity of the mental representations and mechanisms that enable superior performance. Once expert performance is understood as the acquisition of an integrated structure of cognitive and physiological adaptations, then the main challenge is to account for how individual domain experts gradually develop their superior performance during decades of deliberate practice.

Further Reading

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